Bayesian Physics-Informed Neural Networks

Bayesian nonparametrics and Bayesian Machine Learning

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Motivation & Problem Statement

Standard PINNs:

- Learn PDE solutions using limited data and physical laws via deterministic gradient descent.
- **Limitation:** They lack built-in uncertainty quantification and are highly prone to overfitting when sensor data is noisy.

The Solution: Bayesian PINNs (B-PINNs)

- Extend standard PINNs into a Bayesian framework.
- Treat neural network weights probabilistically rather than as fixed values.
- Yields a *distribution* of possible solutions, providing natural uncertainty quantification.

Physics-Informed Neural Networks (PINNs)

General PDE setup:

$$\mathcal{N}_x(u;\lambda) = f \text{ on } D, \quad \mathcal{B}_x(u;\lambda) = b \text{ on } \Gamma$$

where $\mathcal{N}_x$ is a differential operator, λ are PDE parameters, f is the forcing term, and $\mathcal{B}_x$ is the boundary operator.

Observational Data: Dataset $\mathcal{D} = \mathcal{D}_u \cup \mathcal{D}_f \cup \mathcal{D}_b$. Measurements corrupted by Gaussian noise:

$$\bar{u}^{(i)} = u(\boldsymbol{x}_u^{(i)}) + \epsilon_u^{(i)}, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Gaussian prior A Gaussian prior is placed independently on each component of the network weights θ; $P(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$

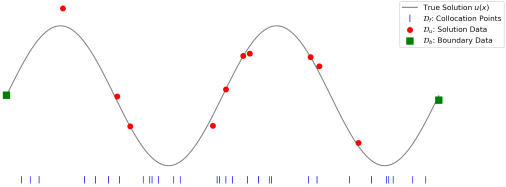


Figure: Example of observational data points sampled from the domain and boundaries.

Posterior by Bayes' theorem:

$$P(\boldsymbol{\theta} \mid \mathcal{D}) \propto \underbrace{P(\mathcal{D} \mid \boldsymbol{\theta})}_{\text{physics-informed likelihood}} \cdot P(\boldsymbol{\theta})$$

Posterior Sampling Approaches

1. Hamiltonian Monte Carlo (HMC)

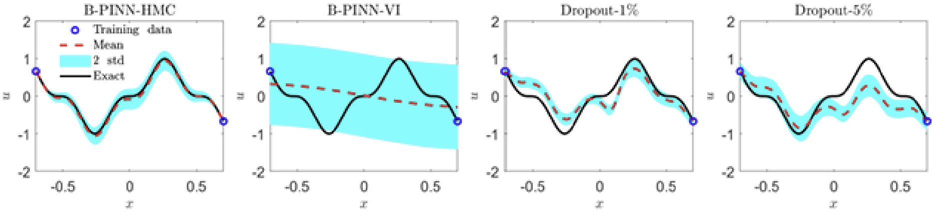
- Introduces auxiliary momentum variable; simulates Hamiltonian dynamics.
- Makes large, informed moves — avoids random-walk behavior.
- **Asymptotically exact** samples from the true posterior.
- Cost: computationally expensive.

2. Variational Inference (VI)

- Recasts inference as optimisation — minimises KL divergence.
- Fits a **fully factorised Gaussian**: each weight assumed independent.
- Fast, but the mean-field assumption **underestimates posterior variance** — overconfident under high noise.

The Dropout is introduced as a cheap baseline.

Key Findings from the Baseline Study



Systematic comparison of standard PINNs, B-PINN (HMC & VI), and Dropout:

- **B-PINN-HMC Excels:** Consistently delivers reliable predictive means and correctly calibrated uncertainty bounds across noise scales.
- **VI Fails:** Factorizable Gaussian assumptions restrict weights from correlating, leading to severely degraded uncertainty quantification.
- **Overfitting Protection:** Under large noise ($\sigma = 0.1$), B-PINN-HMC acts as a regularizer where deterministic PINNs catastrophically overfit.

Metrics for Uncertainty Quantification

Visual checks are insufficient for Bayesian methods. We utilize uncertainty-aware metrics:

- **PICP** (Prediction Interval Coverage): Fraction of true data captured by bounds.
  $\text{PICP} = \frac{1}{N}\sum_{i=1}^{N} c_i, \quad c_i = \mathbb{I}(L_i \leq y_{true,i} \leq U_i)$
- **MPIW** (Mean Prediction Interval Width): Average width of the $\pm 2\sigma$ bands.
  $\text{MPIW} = \frac{1}{N}\sum_{i=1}^{N}(U_i - L_i)$
- **NLL** (Negative Log-Likelihood): Penalizes "confidently wrong" models.

$$\text{NLL} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\log(2\pi\sigma_i^2)}{2} + \frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right)$$

- **ECE** (Expected Calibration Error): Weighted average of bin-wise error.

$$\text{ECE} = \sum_{m=1}^{M}\frac{|B_m|}{N}\left|\text{acc}(B_m) - \text{conf}(B_m)\right|$$

B-PINNs with Unknown Noise

The Unrealistic Assumption:

- Standard models assume the noise level σ_f is perfectly known. In real physics scenarios, this is rarely true.

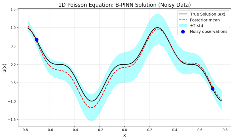
Our Fix: Hierarchical B-PINNs

- We assigned a log-normal prior to the noise and jointly inferred it alongside network weights using HMC.
- **Mathematical Adjustment:** The normalization constant cannot be dropped. The log-likelihood must explicitly retain the variance penalty:

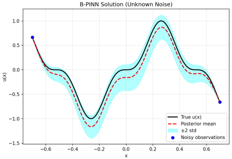
$$\log P(\mathcal{D}_f|\theta, \psi) \propto -N_f\psi - \frac{SSE_f(\theta)}{2\exp(2\psi)}$$

where $\psi = \log \sigma_f$.

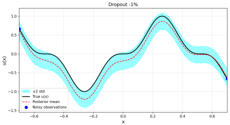
Experimental Results: 1D Poisson equation ($\lambda u_{xx} = f(x)$)



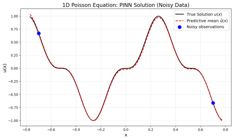
(a) BPINN



(b) BPINN w prior



(c) Dropout 1%



(d)PINN

B-PINN vs. PINN under Noise Regimes

- **High Noise**: Standard PINNs overfit sensor noise $\rightarrow$ high L^2 error.
  B-PINNs treat physics as "soft" constraints, filtering high-frequency noise $\rightarrow$ lower error of 0.2952.
- **Low Noise**: Standard PINNs excel by finding the global minimum of clean data. B-PINNs struggle to sample these "sharp" peaks via HMC, resulting in a slightly higher mean error of 0.2220.

Experimental Results: 1D Poisson equation ($\lambda u_{xx} = f(x)$) (2)

	B-PINN	B-PINN (unknown noise)	PINN	Dropout -5%	Dropout - 1%
Relative L^2 norm	0.2952	0.2251	0.4338	0.3644	0.3713
PICP	92.60	100	-	95.50	54.00
MPIW	0.6465	0.7137	-	0.7864	0.4048
NLL	-0.3885	-0.7345	-	-0.2677	0.3478
ECE	0.1307	0.1016	-	0.0374	0.3090

Table: Metrics for the 1D poisson equation forward problem with measurement noise 0.1

	B-PINN	B-PINN (unknown noise)	PINN	Dropout 5%	Dropout 1%
Relative L^2 norm	0.2220	0.2492	0.0324	0.4341	0.3059
PICP(%)	100	100	-	86.00	95.50
MPIW	0.6389	0.3887	-	0.7398	0.4462
NLL	-0.6452	-0.8415	-	-0.0357	-0.3265
ECE	0.0640	0.2432	-	0.0367	0.1950

Table: Metrics for the 1D poisson equation forward problem with measurement noise 0.01

The behaviour of the B-PINN with a noise prior

- **High Noise**: Superior performance. The inferred noise parameter (σ_f) acts as an adaptive weight, allowing the model to prioritize physical consistency over noisy contradictory data.
- **Low Noise**: Although the model overestimates the 0.01 noise scale, it ensures robustness. This dynamic parameter prevents HMC divergence by avoiding "numerical traps" caused by overly sharp posteriors, maintaining 100% coverage.

Experimental Results: 1D Poisson equation ($\lambda u_{xx} = f(x)$) (3)

	B-PINN	B-PINN (unknown noise)	PINN	Dropout -5%	Dropout - 1%
Relative L^2 norm	0.2952	0.2251	0.4338	0.3644	0.3713
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The impact of the prior on the noise on the uncertainty

- Produces wider, more conservative bands (MPIW 0.7137) at high noise than the vanilla B-PINN
- In low noise, it produces tighter intervals than vanilla B-PINN (0.3887 vs 0.6389), possibly because of a better HMC behaviour due to its inflated noise estimation

Dropout Failure

- Dropout bands are dictated by the *rate*, not the *data*.

Theoretical Motivation: The PI-GP Baseline

Why compare B-PINNs with Gaussian Processes?

- Finite-width B-PINNs completely depend on approximate sampling (HMC, VI) over a highly non-convex posterior energy landscape: $U(\theta) = -\log P(\mathcal{D} \mid \theta) - \log P(\theta)$.
- Sampling depends heavily on hyper-parameter tuning and lacks simple analytical performance guarantees.

Establishing an Absolute Ground-Truth Baseline

- By the Central Limit Theorem, in the infinite-width limit $N_l \to \infty$, a BNN prior converges to a **Neural Network Gaussian Process (NNGP)**: $u(\cdot;\theta) \xrightarrow{\text{Dist.}} \mathcal{GP}\big(0, k_{\text{NNGP}}(\cdot,\cdot)\big)$.
- If the differential operators are **linear**, the B-PINN formulation analytically reduces to a **Physics-Informed Gaussian Process (PI-GP)**.
- The PI-GP bypasses sampling, allowing us to strictly benchmark the adaptivity of finite-width B-PINNs against exactly the mathematical limit they are derived from.

The Infinite-Width Limit: BNNs as NNGPs

From Neural Networks to Gaussian Processes:

- Consider a fully-connected BNN surrogate $\tilde{u}(x;\theta)$ with hidden layers of width N_l.
- By the Central Limit Theorem, as the hidden layer widths $N_l \to \infty$, the pre-activations converge to a multivariate Gaussian.
- Consequently, the **output surrogate function converges to a Gaussian Process**:

$$u(\cdot;\theta) \xrightarrow[N_l \to \infty]{\text{Distribution}} \mathcal{GP}\big(0, k_{\text{NNGP}}(\cdot,\cdot)\big)$$

The Kernel k_{NNGP}:

- The prior covariance kernel is recursively determined by the network depth, variance priors (σ_w^2, σ_b^2), and the activation function ϕ:

$$K^{(0)}(x,x') = \frac{\sigma_w^2}{d_{in}} x \cdot x' + \sigma_b^2 \qquad K^{(l)}(x,x') = \sigma_w^2 \mathbb{E}_{(z,z')\sim\mathcal{N}(0,\Sigma^{(l-1)})}\Big[\phi(z)\phi(z')\Big] + \sigma_b^2$$

- **Implication:** An infinitely wide B-PINN prior induces an exact GP prior on the predicted solution field $u(\cdot)$.

Gaussian Processes under Linear Differential Operators

The exact equations defining the PI-GP block Covariance K:

- If the differential operator $\mathcal{L}_x$ is **linear**, applying it to $u \sim \mathcal{GP}\big(0, k_{\text{NNGP}}(x, x')\big)$ yields a Gaussian Process. The spatial operators commute with the covariance expectation:

$$\text{Cov}\big(u(x), \mathcal{L}_{x'} u(x')\big) = \mathcal{L}_{x'} k_{\text{NNGP}}(x, x')$$

$$\text{Cov}\big(\mathcal{L}_x u(x), \mathcal{L}_{x'} u(x')\big) = \mathcal{L}_x \mathcal{L}_{x'} k_{\text{NNGP}}(x, x')$$

Analytical PI-GP Posterior:

- With data $\mathcal{D}$, the inference reduces to a **closed-form GP regression**:

$$\mu(x^*) = \mathbf{k}_*(\mathbf{K} + \Sigma_{\text{noise}})^{-1} Y$$
$$\Sigma(x^*) = k(x^*, x^*) - \mathbf{k}_*(\mathbf{K} + \Sigma_{\text{noise}})^{-1} \mathbf{k}_*^T$$

- No more iterative neural network training and sometimes unstable MC sampling, yielding an exact, resolution-free, convex analytical ground-truth posterior.

Empirical Evaluation: Expressivity vs. Adaptivity

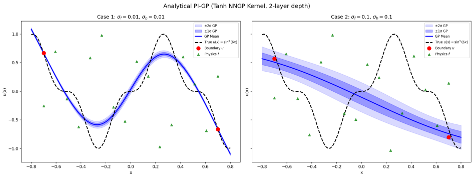

Figure: PI-GP posterior mean against truth $u(x) = \sin^3(6x)$ for low and high noise.

Failure of the PI-GP vs. Finite B-PINNs:

- **Too Rigid:** The infinite-width limit forces the network to a "lazy regime" of NTK.
- **Smoothing effect:** This rigid structure causes the PI-GP to heavily over-smooth the physics in the presence of noise, missing the fast oscillating details of the wave.
- **Finite Networks Adapt:** Finite-width B-PINNs do not have fixed kernels. Their basis functions actively shift and adapt during training to perfectly align with sharp gradients and complex PDE dynamics.

Conclusion

- **B-PINNs $>$ PINNs for Noise:** HMC-driven B-PINNs successfully regularize against overfitting in chaotic PDE problems.
- **Metrics Matter:** Rigorous quantitative evaluation (PICP, MPIW) is crucial—visuals and Dropout are deceptive.
- **Dynamic Noise is Powerful:** Joint inference of noise prevents numerical divergence and improves HMC sampling stability.
- **Finite vs Infinite (PI-GP):** While mathematically elegant, infinite-width exact PI-GPs fail on complex dynamics because their rigid kernels over-smooth. Finite-width B-PINNs are essential because their active representation learning can perfectly adapt to sharp features.

Future Research

- **Unknown noise in inverse problems:** Validating whether the unknown noise formulation successfully regularizes inverse problems without sacrificing PDE parameter identifiability.
- **Stochastic Gradient MCMC for Scalability:** Investigating SG-MCMC methods to enable mini-batching, reducing computational cost while preserving asymptotic convergence.
- **Active Learning & Optimal Sensor Placement:** Leveraging predictive posterior variance to guide sensor placement and collocation points, maximizing information gain while minimizing data acquisition costs.
- **Adaptive Priors and Architectures:** Exploring the influence of different weight priors and network architectures to improve regression of multi-scale phenomena.